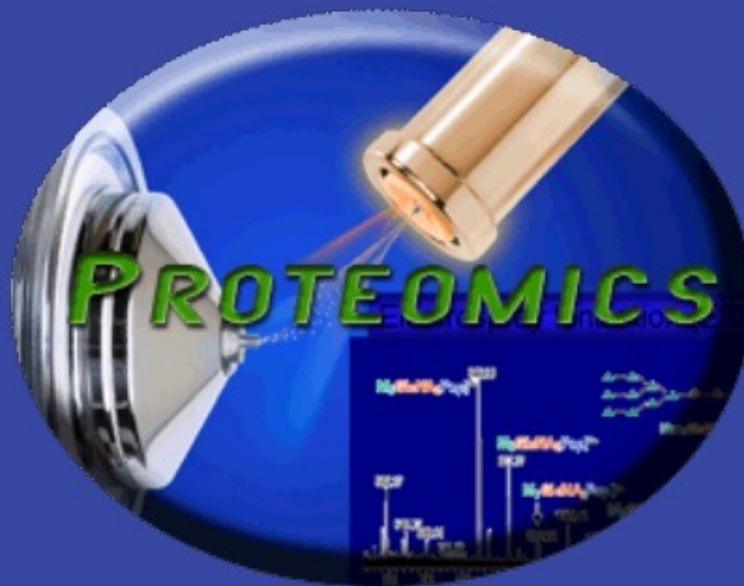


Proteomics: Protein Profiling and Identification through Mass Spectrometry



Vermont Genetics Network
Proteomics Outreach
Program

What is an IDeA State?



The National Institutes of Health (NIH) Institutional Development Award Program (IDeA) was established in 1993 to broaden the geographic distribution of NIH funding for biomedical and behavioral research.

- Increase Competitiveness
- Build capacity
- Advance research through animal models
- Foster research through technologies
- Develop informatics approaches to support research
- Strengthen the research workforce
- Maximize partnerships



Vermont Genetics Network (VGN)

Founded at the University of Vermont in 2001 through an NIH BRIN grant and renewed in 2005 through an NIH INBRE grant

- **Encourage biomedical research in Vermont**
- **Create a “network” of researchers and students**
- **Give outreach lectures to 4-year institutions**
- **Provide research grants to junior faculty and students**
- **Mentoring for students interested in research**
- **Core Facility Development and Support**



Vermont Genetics Network

IDeA Networks of Biomedical Research Excellence



INBRE: Vermont Genetics Network Core Facilities

Microarray Core

-Established in 2002

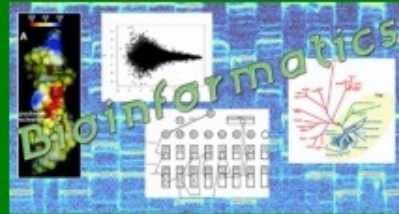


UNIVERSITY OF VERMONT
Microarray Facility



Bioinformatics Core

-Established in 2002



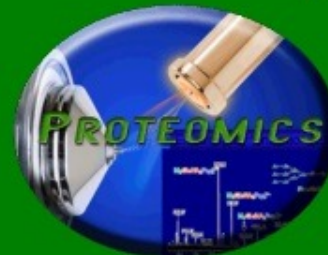
Outreach Core

-Established in 2003



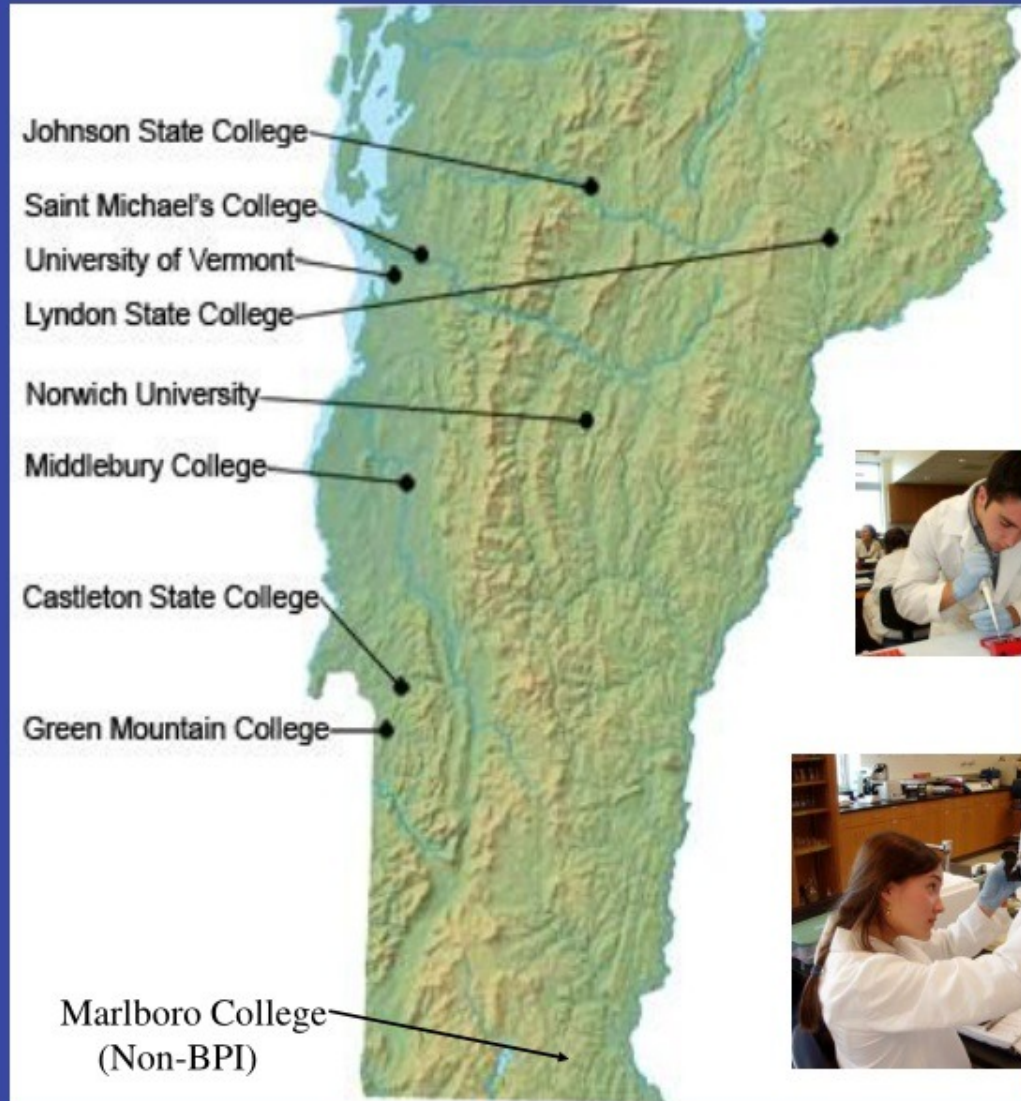
Proteomics Core

– Established in 2006



Building Statewide Network Through Outreach

Activities: Inter-Institutional



2001: 5 BPI Partners

-Develop culture of research

-Funding for Faculty Development and Undergraduate Student Research

-Infrastructure Support

-Career Development for Undergraduate Students

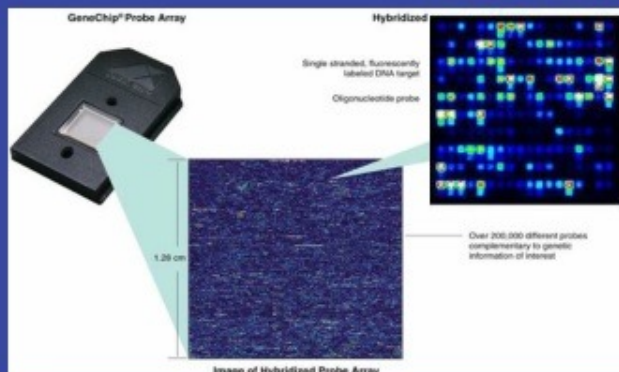
-Access to UVM Core Facilities and Expertise

2010 submission: 7 BPI partners

VGN Outreach Modules:

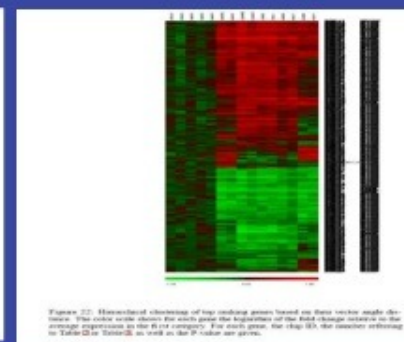
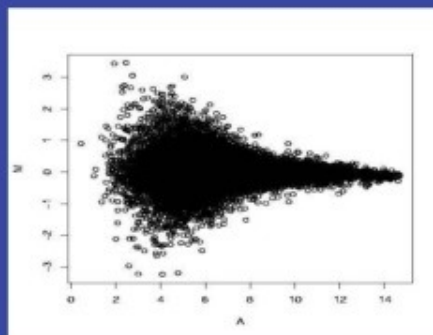
- Microarray

First Delivery in 2003



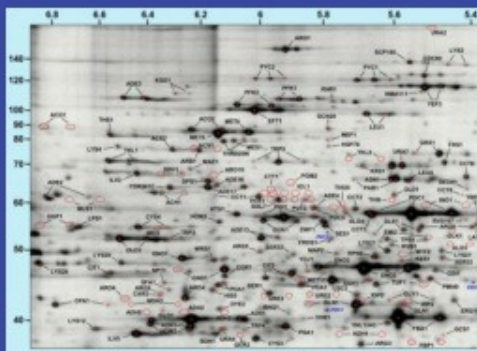
- Bioinformatics

First Delivery in 2006



- Proteomics

First Delivery in 2009



Sites that have participated in Microarray Outreach:

UVM

St. Michael's College

Johnson State College

Middlebury College

Green Mountain College

Norwich University

Castleton State College

Lyndon State College

Marlboro College

Microarray Outreach Outcomes:

The coordination and delivery of the microarray modules to each Vermont college is only the beginning of our relationship with the college's faculty and students.

Core :

- Collaborations
- Facility Usage
- Networking Throughout the State
- Interactions with Students

Faculty:

- Gaining Expertise
- Curriculum Development
- Granting Opportunities
- Collaborations
- More Scientific Colleague

Undergraduate Students:

- Exposure to Cutting Edge Technology
- Interactions with Faculty and Staff at UVM
- Career Networking
- Internships
- Summer Jobs

Proteomics Outcomes...These and More???

VGN Proteomic Outreach Program

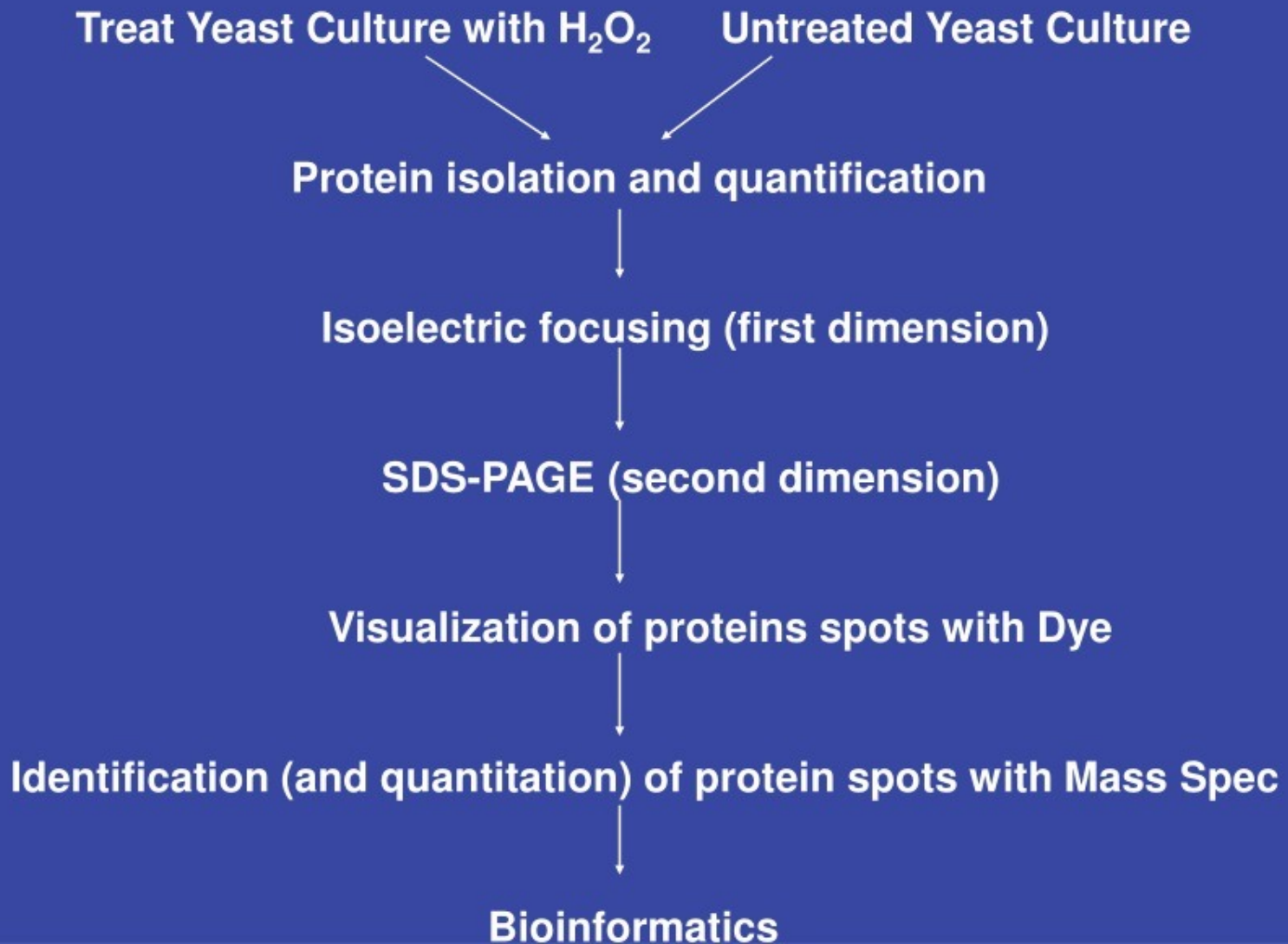
We provide all the equipment, supplies and teaching materials.

The manual, references and ppt. presentations are all available online.

Goal is to integrate the proteomics module into the college's curriculum.

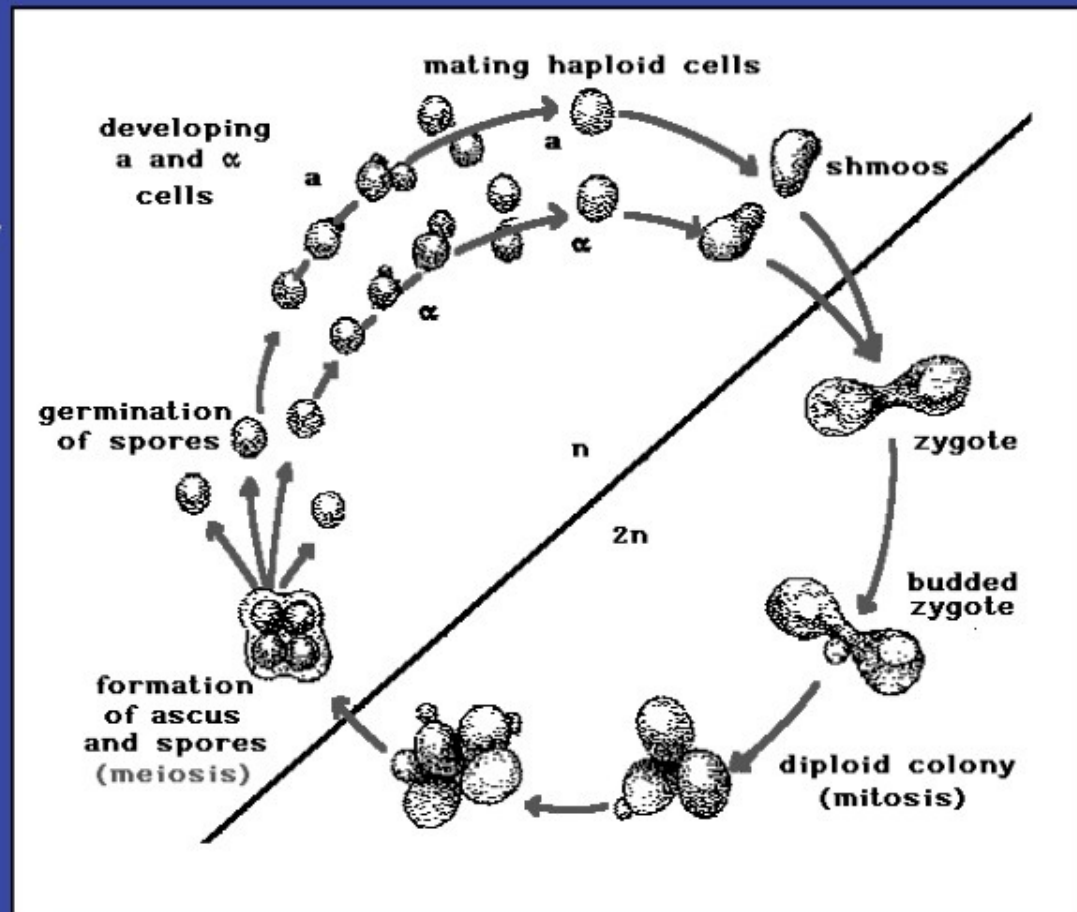
Experiment and Workflow:

-The effect of oxidative stress (using chemical Hydrogen Peroxide (H_2O_2) on protein expression in yeast *S. cerevisiae*



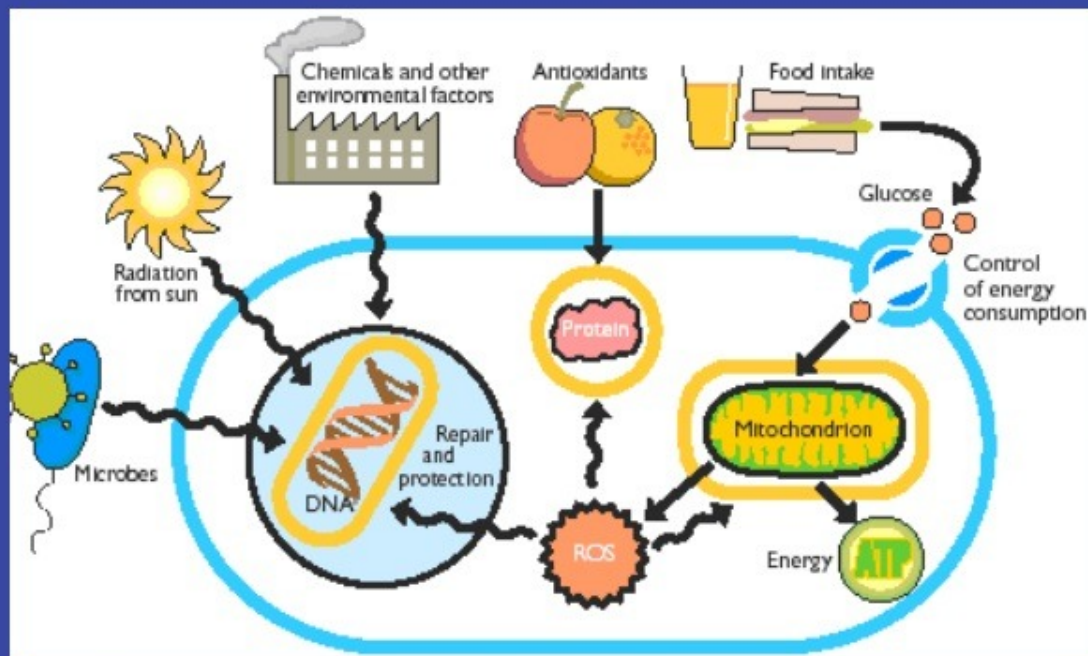
Why Use Yeast (*S. cerevisiae*)??

- easily manipulated in the laboratory
- simple eukaryote, unicellular
- rapid growth (doubling 1.5 - 2.5 hours)
- non-pathogenic
- stable haploid and diploid states
- complete genome sequenced



What causes Oxidative Stress?

When the concentration of ROS present in the cell exceeds the capacity of the cells ability to detoxify or to repair damages



Role of H_2O_2 :

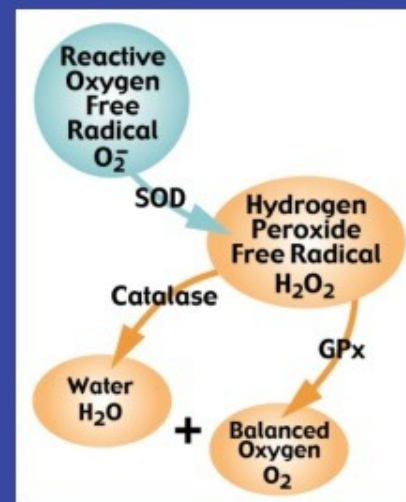
-Hydrogen Peroxide is formed from an oxygen radical and considered to be an ROS because it has the ability to form $\cdot OH$ in the presence of metal ions

-Defense Mechanism in *Saccharomyces cerevisiae*

Oxidative Stress: H_2O_2 Overload

This occurs when the capacity of Catalase and Glutathione peroxidases, it can be reduced to form $\cdot OH$

The hydroxyl radical is highly reactive and can lead to:
DNA Degradation
Protein Peroxidation
Lipid Peroxidation



Yeast Proteomics Experiment



untreated



versus



H₂O₂ treated

Changes in protein expression?

Are new proteins identified?

What do the results say about yeast biology?

What parallels (if any) can we make to human biology?

Good Luck with the experiment!



VGN Outreach Instructors:

**Pat Reed
Scott Tighe
Bryan Ballif
Janet Murray
Tim Hunter**

